

Edge-Aware Multi-Modal Intrusion Detection for Consumer-Centric Industrial Internet-of-Things Devices Using Deep Adaptive Fusion

Nimra Nasir, Amnah Firdous, Syeda Sitara Waseem, Syed Rizwan Hassan, Mansoor Ihsan, Isma Farah Siddiqui

Abstract—The rapid proliferation of Consumer-Centric Industrial Internet-of-Things (CIIoT) ecosystems has substantially expanded the attack surface for diverse cyber threats. Conventional intrusion detection systems (IDS) primarily rely on signature-based techniques, which exhibit limited effectiveness against zero-day and evolving attack patterns. To address these limitations, this study proposes a deep learning-driven IDS framework that concurrently integrates multiple modality-specific models. The proposed architecture employs a Gated Recurrent Unit (GRU) for host system log analysis, a Long Short-Term Memory (LSTM) network for network flow modeling, a 1D Transformer for packet-level traffic inspection, and a fine-tuned TinyBERT model for phishing email detection. The probabilistic outputs of these heterogeneous models are combined through a learnable adaptive fusion mechanism, enabling confidence-aware decision aggregation across modalities. Extensive experiments conducted on the UNSW-NB15, CIC-IDS2018, CTDAPD, and phishing email datasets demonstrate that the proposed framework outperforms conventional IDS approaches, achieving improved accuracy, F1-score, and ROC-AUC while significantly reducing false positive rates. The framework is explicitly designed as edge-aware through lightweight architectures edge-aware and late-fusion strategies, while detailed deployment benchmarking is reserved for future work.

Index Terms—Consumer-Centric Industrial Internet-of-Things (CIIoT), Intrusion Detection, LSTM, TinyBERT, Cyber-Attacks.

I. INTRODUCTION

THE exponential growth of interconnected smart nodes in Internet-of-Things (IoT) environments has substantially increased vulnerability to sophisticated cyber threats. Malicious activities, including system intrusions, phishing operations, malware propagation, and service disruption attacks, compromise sensitive information, interrupt operational continuity, and endanger critical infrastructure components [1], [2]. Traditional intrusion detection mechanisms frequently

prove inadequate against emerging cyber threats due to their dependence on signature-based recognition and limited generalization capacity [3].

Cybersecurity breaches can produce severe consequences, including data exposure, service unavailability, and reputational harm. In industrial and healthcare contexts, undetected intrusions can potentially endanger human safety. The escalating complexity of attack methodologies, combined with imbalanced datasets, further complicates detection efforts and typically generates excessive false positive rates [4]–[6].

Contemporary research has focused on deep learning-based IDS solutions to address these challenges. Sequential models, autoencoders, and transformer architectures have demonstrated proficiency in learning temporal and spatial relationships within network traffic patterns [7], [8]. Hybrid approaches, including feature fusion, meta-learning, and transfer learning, have shown promising improvements in robustness and detection accuracy [9]–[11]. Explainable AI (XAI) methodologies have also been incorporated to enhance decision transparency and interpretability [12], [13]. Despite these advancements, current systems continue to face challenges, including suboptimal performance on imbalanced datasets, insufficient cross-dataset generalization, and inadequate model explainability.

This investigation presents a fusion-based deep learning (DL) intrusion detection framework that combines sequence-oriented (GRU/LSTM) and transformer-based architectures to leverage complementary learning characteristics. Computationally efficient feature engineering and late fusion strategies are implemented to improve detection performance across diverse publicly available datasets, including network traffic and phishing email collections [14], [15]. The system additionally incorporates explainability components through ROC-AUC analysis and attention-based visualization to enhance practical deployment viability.

A. Motivation and Contribution

The growth of Consumer-centric Industrial IoT (CIIoT) networks has heightened vulnerability to intrusions, phishing, and denial-of-service (DoS) attacks. Traditional signature-based intrusion detection systems (IDS) are often ineffective against novel attack strategies and face challenges like class imbalance, scarce labeled data, and poor interpretability. While DL offers promise by learning hierarchical and temporal patterns, single-model architectures struggle to generalize across diverse data types. To overcome these issues, this study introduces a multimodal DL framework that combines sequence

Nimra Nasira and Syeda Sitara Waseema with are the Department of Computer Science & IT, The Govt. Sadiq College Women University, Bahawalpur, 63100, Pakistan.(Email: nimranasir122@gmail.com, syeda-sitara14@gmail.com)

Amnah Firdous is with the Department of Computer Science, The Islamia University of Bahawalpur, Bahawalpur, 63100, Pakistan. (Email: amnah@gscwu.edu.pk)

Syed Rizwan Hassan is with the Department of Computer Engineering, Gachon University, Seongnam-si, South Korea. (Email:syed9919@gachon.ac.kr)

Mansoor Ihsan is with the The University of Salford, Manchester, UK. (Email:enquiries@greatermanchester.ac.uk)

Isma Farah Siddiqui is Department of Software, Systems and Cybersecurity, Faculty of Information Technology, Monash University, Victoria, Australia. (Email: Ismafarah.siddiqui@monash.edu)

(Corresponding author: Syed Rizwan Hassan)

models (LSTM, GRU) with transformer-based architectures (1D Transformer, TinyBERT) for robust and interpretable intrusion detection across heterogeneous datasets. The primary contributions of this work are summarized as follows:

- A novel hybrid GRU–Transformer fusion architecture designed for multimodal intrusion detection scenarios
- Implementation of advanced feature engineering and data balancing techniques to enhance robustness and detection performance
- Comprehensive experimental evaluation across heterogeneous datasets including UNSW-NB15, CIC-IDS2018, CTAP host logs, and phishing email data
- This study provides preliminary interpretability analysis using attention-based representations and ROC-AUC visualization transparent decision-making
- Demonstrated superiority over existing IDS models through improved accuracy and reduced false positive rates

Existing ensemble intrusion detection systems often use static fusion methods like majority voting or averaging that treat all data types as equally reliable. However, in CCIIoT environments, security data is inherently varied, including network flows, logs, and text, each with distinct patterns. Fixed fusion fails to adapt when some sources become noisy or less informative, leading to suboptimal detection. To overcome this, we propose an adaptive fusion mechanism that learns modality-specific importance weights, dynamically integrating complementary signals while suppressing unreliable ones. Notably, to our knowledge, no current CCIIoT framework employs such a learnable, adaptive approach to combine sequence models, transformers, and language models across network, host, and textual data. Our work bridges this gap by introducing a deep adaptive fusion framework to improve robustness, generalization, and accuracy in heterogeneous CCIIoT settings.

B. Paper Structure

The remainder of this paper is organized as follows. Section II reviews relevant literature on DL-based intrusion detection. Section III details the data collection, preprocessing, and model configuration methodology. Section IV elaborates on individual model architectures and the fusion mechanism. Section V presents experimental results and performance analysis. Finally, Section VI concludes the paper and suggests potential future research directions.

II. RELATED WORK

Artificial intelligence and DL are widely applied to enhance network intrusion detection systems. Research focuses on hybrid models, feature fusion, and lightweight designs to boost accuracy and efficiency. Kaur et al. [1] highlight the promise of hybrid and ensemble methods for advanced threats, while Thakkar and Lohiya [16] improve detection by integrating statistical feature selection in DNN-based IDS, reducing computational costs. To address scarce labeled attack data, meta-learning and few-shot techniques have been explored. Lu et al. [11] propose a meta-learning framework for CCIIoT

intrusion detection, and Zhang et al. [10] enhance malware traffic classification via knowledge transfer and architecture search. Deep sequence models like LSTM and GRU have proven effective for capturing temporal dependencies in traffic [7], [8].

For resource-constrained CCIIoT and edge environments, lightweight supervised and hybrid schemes have been developed. Roy et al. [3] present a low-latency IDS for CCIIoT, and Thakkar et al. [16] combine dimensionality reduction with LSTM for enhanced performance. Alzubi et al. [17] design a hybrid DL framework for edge IoMT, and Louk and Tama [18] demonstrate robust anomaly detection using ensemble gradient boosting. Recent advancements in consumer electronics security have achieved notable progress. Alzubi et al. [17] created a blended DL model for edge-based IoMT, enabling efficient intrusion detection for home medical devices. Chen et al. [19] proposed a lightweight CNN optimized for smart home devices, achieving high accuracy with minimal resources. Zhang and Li [20] introduced a federated learning framework for consumer CCIIoT that preserves privacy through collaborative edge-device training.

Additional contributions address explainability: [21] edge-based processing [22] and multi-modal fusion for smart cameras [23] collectively demonstrating a trend toward adaptive, explainable, and efficient DL frameworks. Collectively, these studies demonstrate a trend toward adaptive, explainable, and efficient DL frameworks that leverage multimodal fusion, privacy-aware learning, and edge computing for advanced intrusion detection Table I.

Current fusion-based IDSs rely on static aggregation (e.g., voting, averaging), assuming equal reliability across all modalities. However, CCIIoT data is heterogeneous network flows, logs, packets, and text; each has distinct statistical properties and noise levels. Static methods cannot adapt to varying data quality and underperform with sparse or noisy sources. Moreover, existing research focuses on single datasets or modalities, limiting cross-domain generalization. To our knowledge, no CCIIoT framework employs a learnable, modality-aware adaptive fusion mechanism that integrates sequence models (LSTM/GRU), transformers, and language models (TinyBERT) across network, host, and textual data. This gap motivates our deep adaptive fusion framework, which dynamically aggregates heterogeneous outputs to improve accuracy, robustness, and generalization in CCIIoT.

III. PROPOSED METHODOLOGY

This research employs a multi-model DL approach for cybersecurity threat identification and classification from diverse data sources. We utilized four distinct datasets, each processed by a specialized DL model according to its data characteristics. Figure 1 illustrates the methodological pipeline encompassing data collection, preprocessing, model construction, training, and evaluation stages. All modality-specific models operate in parallel on their respective data streams, and their probabilistic outputs are subsequently integrated through the proposed adaptive fusion module.

TABLE I: Summary of Key Related Works on Deep Learning for Intrusion Detection

Ref.	Approach / Model	Dataset(s)	Key Findings / Contribution
[1]	AI/ML cybersecurity review	Multiple	Identified hybrid models as key for advanced threats
[3]	Lightweight supervised model	CCIoT Networks	Resource-efficient detection for CCIoT devices
[7]	Sequential DL (LSTM/GRU)	UNSW-NB15	Captured temporal dependencies for sequential attacks
[9]	DNN with statistical fusion	UNSW-NB15	Statistical feature selection improved accuracy
[11]	Few-shot meta-learning	CCIoT Traffic	Effective detection with limited labeled data
[10]	Knowledge transfer + NAS	Malware Traffic	Enhanced few-shot malware classification
[24]	Enhanced component analysis	WSN Data	Improved WSN intrusion detection
[25]	Hierarchical multi-stage	NIMS	Progressive filtering improved reliability
[16]	LSTM with feature reduction	UNSW-NB15	Dimensionality reduction enhanced feature representation
[17]	Blended DL framework	Edge-IoMT Data	Edge-centric framework for IoMT industries
[18]	Bagging-based GBDT	Multiple datasets	Ensemble improved robustness against diverse attacks
[26]	Sin-Cos-BIAVOA feature selection	DDoS-specific	Novel optimization for DDoS feature selection
[27]	End-to-end DL	Industrial CCIoT	Learned features directly from raw data
[28]	Bio-inspired hybrid DL	NSL-KDD, UNSW-NB15	Biological mechanisms for high accuracy

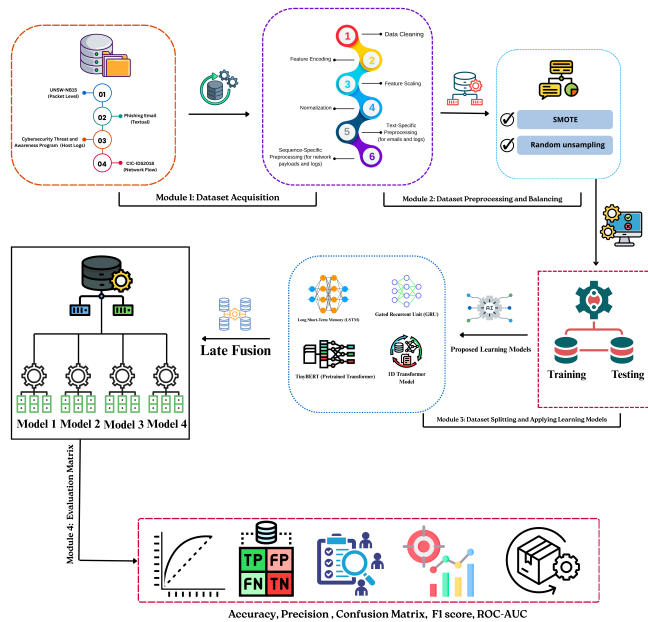


Fig. 1: Architecture of Multimodal Intrusion Detection

A. Data Collection

This study utilizes four heterogeneous datasets representing host logs, packet-level data, network flows, and textual emails to capture diverse cybersecurity threats Table II. All data are preprocessed and labeled under two categories: Normal and Attack.

1) *Host Logs Dataset (CTAP)*: This dataset [29] contains simulated system event logs reflecting real world host-level activities, including authentication details, file access, and process execution critical for detecting insider threats and anomalous host behaviors.

2) *UNSW-NB15 Dataset (Packet Level)*: This dataset [14] includes raw packet captures with both normal and malicious traffic. For our study, 43 flow-based features were used, excluding redundant identifiers (e.g., IPs). Attack labels were mapped to a binary setting (Normal vs Attack).

3) *CIC-IDS2018 Dataset (Network Flow)*: The CIC-IDS2018 dataset [30] provides realistic network traffic flows

with comprehensive flow-based features, effective for anomaly and intrusion detection.

4) *Phishing Email Dataset (Textual)*: This dataset [15] consists of email metadata and content-based attributes for detecting malicious and phishing messages, capturing lexical, structural, and header-based characteristics.

TABLE II: Attributes of Host Log and Packet-Level Datasets

Host Log Dataset	Packet-Level Dataset
<i>Timestamp</i> : Log generation time	<i>Source IP</i> : IP address of sender
<i>Consumer ID</i> : Identifier of the user	<i>Destination IP</i> : IP address of receiver
<i>Event Type</i> : Activity type (login, file access)	<i>Source Port</i> : Port number of sender
<i>Process Name</i> : Name of executed process	<i>Destination Port</i> : Port number of receiver
<i>File Accessed</i> : File involved in event	<i>Protocol</i> : TCP, UDP, ICMP
<i>Logon Status</i> : Failure of login attempt	<i>Packet Size</i> : bytes
	<i>Timestamp</i> : Time of packet capture

These four datasets collectively represent CCIoT's heterogeneous landscape: (1) CTAP host logs capture device-level anomalies; (2) UNSW-NB15 enables packet-level threat inspection; (3) CIC-IDS2018 provides scalable flow-based analysis; (4) Phishing emails cover human-centric textual attacks. Together, they comprehensively span network, host, packet, and human interaction layers characteristic of consumer-centric IIoT environments.

B. Data Preprocessing

Following data acquisition, raw datasets undergo comprehensive preprocessing to ensure data consistency and model readiness. Missing values are addressed through imputation (mean/median for numerical, mode for categorical), and duplicate records are removed to prevent bias. Categorical variables such as protocol types and service properties are encoded via one-hot or ordinal encoding, and all numerical attributes are normalized to a [0,1] range using Min-Max normalization to avoid feature scale bias.

Textual data, including phishing emails and logs, is tokenized, cleaned of stop-words, and lowercased. Network

payloads undergo duplicate symbol elimination and are padded/truncated to fixed sequence lengths, while system logs are standardized by timestamp, filtered for relevance, and transformed into structured numerical sequences suitable for model input.

C. Data Balancing

Class imbalance, where benign instances vastly outnumber attacks, causes model overfitting to majority classes and poor rare-attack detection. We address this using SMOTE (generating synthetic minority samples via interpolation) and random undersampling (reducing majority class size while preserving key patterns). These techniques collectively balance class distributions and enhance detection of both common and rare attacks.

D. Experimental Protocol and Data Integrity

A strict data protocol prevents leakage. Datasets are split 80/10/10 via stratified sampling, with no sample overlap. SMOTE/undersampling is applied to the training set only; validation/test sets remain original. Feature scaling uses training-set parameters exclusively. For time-series data, time-aware splitting ensures training precedes validation/testing, preventing temporal leakage. All metrics reported reflect a true generalization without contamination. To ensure reproducibility, all experiments used fixed random seeds (42 for Python/NumPy, 42 for TensorFlow).

IV. MODEL CONFIGURATION

A. Model 1: Long Short-Term Memory (LSTM) Model

LSTM networks address the vanishing gradient problem through memory cells and gating mechanisms. This architecture effectively captures long-range temporal dependencies, which are crucial for network intrusion detection [31]. At each timestep t , the LSTM unit updates its hidden state h_t and cell state c_t using three gating components: forget, input, and output. The operations are mathematically defined as:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f), \quad (1)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i), \quad (2)$$

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c), \quad (3)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \quad (4)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o), \quad (5)$$

$$h_t = o_t \odot \tanh(c_t), \quad (6)$$

where $\sigma(\cdot)$ denotes the sigmoid function, $\tanh(\cdot)$ represents the hyperbolic tangent, and $\odot$ indicates element-wise multiplication. The CIC-IDS2018 dataset is modeled as sequences of feature vectors, where each sequence corresponds to a network flow. The following architecture, illustrated in Figure 2, captures temporal dependencies in traffic behavior:

- 1) **Input Layer:** Input comprises sequences with dimensionality (T, F) , where T represents timesteps and F denotes features. Variable-length sequences undergo zero-padding.

- 2) **Masking Layer:** A masking layer ignores padded positions with zero values, ensuring padding does not affect training.
- 3) **LSTM Layer:** An LSTM layer containing 64 hidden units processes sequences. With `return_sequences` set to `False`, the layer outputs only the final timestep hidden state $h_T \in \mathbb{R}^{64}$, serving as comprehensive sequence representation.
- 4) **Dense Embedding Layer:** The hidden state projects into a higher-dimensional latent representation using a fully connected layer with 128 units:

$$z = W_e h_T + b_e, \quad z \in \mathbb{R}^{128}. \quad (7)$$

- 5) **Output Layer:** A single neuron dense layer with sigmoid activation performs binary classification:

$$\hat{y} = \sigma(W_o z + b_o). \quad (8)$$

The model trains using binary cross-entropy loss function [32]:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right], \quad (9)$$

where $y_i \in \{0, 1\}$ represents ground-truth class labels and $\hat{y}_i$ indicates predicted probabilities.

The LSTM architecture defined by Equations 1–6 enables effective learning of temporal patterns in network traffic flows, as shown in the model schematic Figure 2. The embedding transformation in Equation 7 creates rich feature representations, while the final classification output in Equation 8 provides intrusion detection probabilities. The optimization objective in Equation 9 ensures the model learns to distinguish between normal and malicious network activities. Hyperparameter for this model were selected using grid search on the validation set to balance detection performance and computational efficiency.

B. Model 2: Gated Recurrent Unit (GRU)

This model processes sequential host log data for intrusion detection [31] using stacked Bidirectional GRUs with an attention mechanism, with the overall architecture depicted in Figure 2.

The GRU model exhibits lower standalone performance due to severe class imbalance in CTAP (Host Logs) and their sparse, repetitive semantic content. However, it provides complementary behavioral signals (e.g., authentication events, process execution) not captured by network or packet data. The proposed fusion mechanism automatically down-weights low-confidence GRU outputs, preventing performance degradation while preserving multimodal diversity.

- 1) **Bidirectional GRU Layers:** The architecture uses two stacked Bidirectional GRU layers, each with 64 units see

Figure 2. The GRU cell updates its hidden state h_t at timestep t through gating operations:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (\text{Update Gate}) \quad (10)$$

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (\text{Reset Gate}) \quad (11)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (12)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (13)$$

The fusion weights α_i are sample-adaptive: computed per instance from modality-specific confidence signals. During training, fusion parameters $\{w_i\}$ are jointly optimized via validation-set gradients while base models remain frozen, enabling dynamic weighting that adapts to varying data quality across modalities.

2) *Attention Layer*: An attention mechanism focuses on critical host events. For each timestep, the attention score α_t and context vector c are computed:

$$e_t = \tanh(W_a h_t + b_a), \quad (14)$$

$$\alpha_t = \frac{e^t}{\sum_{k=1}^T e^{e_k}}, \quad (15)$$

$$c = \sum_{t=1}^T \alpha_t h_t \quad (16)$$

Hyperparameters were selected using grid search on the validation set to balance detection performance and computational efficiency.

C. Model 3: 1D Transformer Encoder for UNSW-NB15

This model uses a one-dimensional Transformer Encoder [31] to capture both local and long-range dependencies in packet flow sequences from the UNSW-NB15 dataset. The encoder-based architecture is illustrated in Figure 2.

1) *Transformer Encoder Blocks*: The encoder processes input sequences through stacked blocks Figure 2. Each block first applies Multi-Head Self-Attention (MHSA) to contextualize token representations, followed by a feed-forward network for non-linear transformation. Residual connections and layer normalization are used after each operation:

$$\text{MHA}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (17)$$

$$H_1 = \text{LayerNorm}(X' + \text{MHA}(X')), \quad (18)$$

$$H_2 = \text{LayerNorm}(H_1 + \text{MHA}(H_1)), \quad (19)$$

$$H_2 = \text{LayerNorm}(H_2 + \text{FFN}(H_2)), \quad (20)$$

$$\text{FFN}(x) = \max(0, xW_1 + b_1)W_2 + b_2. \quad (21)$$

2) *Pooling Layer*: A fixed-dimensional representation for classification is created by pooling across the sequence dimension:

$$h_{\text{mean}} = \frac{1}{T} \sum_{t=1}^T H_{\text{norm},t}, \quad (22)$$

$$h_{\text{max}} = \max_{t=1}^T H_{\text{norm},t}, \quad (23)$$

$$h_{\text{pool}} = [h_{\text{mean}} \parallel h_{\text{max}}]. \quad (24)$$

This pooled vector is passed to a final classifier. Hyperparameters were selected via grid search on the validation set to balance performance and computational efficiency.

D. Model 4: TinyBERT for Phishing Email Detection

The fourth component uses TinyBERT, a distilled BERT variant offering efficient yet capable linguistic representation [31], making it suitable for resource-constrained environments requiring semantic understanding. The Transformer encoder architecture underlying TinyBERT follows the principles shown in Figure 2.

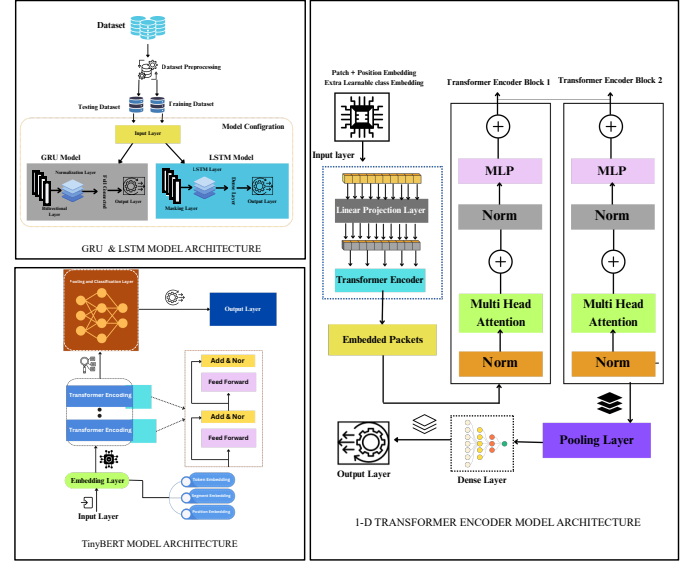


Fig. 2: Evaluated model architectures: (1) GRU with masking and normalization, (2) LSTM, and (3) Transformer Encoder with multi-head self-attention.

1) *Architecture and Adaptation*: TinyBERT builds on the Transformer encoder architecture Figure 2. The core self-attention mechanism is scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (25)$$

where scaling prevents gradient vanishing.

Multi-head attention captures diverse contextual relationships:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V), \quad (26)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O. \quad (27)$$

For classification, the final hidden state of the '[CLS]' token is passed to:

$$\hat{y} = \text{softmax}(W_{\text{cls}} \cdot H_{[\text{CLS}]} + b_{\text{cls}}). \quad (28)$$

2) *Model Configuration and Training*: We use the TinyBERT_General_4L_312D variant with 4 layers and 312 hidden size. The model is fine-tuned on our phishing email dataset with hyperparameters in Table III. A linear learning rate warmup stabilizes early training over the first 6% of steps. Hyperparameters were tuned via validation-set grid search to balance accuracy and efficiency.

TABLE III: Hyperparameter Configurations for All Models

Parameter	LSTM	GRU	1D Trans.	TinyBERT
Architecture	LSTM	GRU	Transformer	TinyBERT
Hidden Units	64	128	—	—
Heads/Layers	1/1	1/1	4/2	4/4
Embed Dim	128	—	Variable	312
Seq Length	Variable	Variable	Variable	64
Optimizer	Adam	Adam	AdamW	AdamW
LR	0.001	0.001	3×10^{-4}	2.5×10^{-5}
Batch Size	64	64	64	16–32
Epochs	10	50	30	3–6
Loss	Binary CE	Categorical CE	Binary CE	Cross-Entropy
Dropout	—	0.3	0.1	0.1
Weight Decay	—	—	1×10^{-4}	0.01

E. Deep Adaptive Fusion Mechanism

Effective intrusion detection requires unified decision-making that accounts for the varying reliability of the modality. We introduce a Deep Adaptive Fusion mechanism to dynamically integrate outputs from all four models in a confidence-aware manner.

Let $\mathcal{M} = \{M_1, M_2, M_3, M_4\}$ be the trained models respectively, the GRU host log analyzer, LSTM network flow model, 1D Transformer packet inspector, and TinyBERT phishing detector. For an input sample, each outputs a probability:

$$p_i = M_i(x_i), \quad p_i \in [0, 1], \quad i \in \{1, 2, 3, 4\} \quad (29)$$

The fusion module learns adaptive importance weights α_i via a fully connected layer and softmax:

$$\alpha_i = \frac{\exp(w_i)}{\sum_{j=1}^4 \exp(w_j)}, \quad \sum_{i=1}^4 \alpha_i = 1 \quad (30)$$

The final intrusion probability is the weighted aggregation:

$$P_{\text{final}} = \sum_{i=1}^4 \alpha_i \cdot p_i \quad (31)$$

Parameters are trained jointly via validation-driven optimization, allowing the model to emphasize modalities with higher confidence or discriminative power per sample. This strategy ensures robustness against noisy, unbalanced, or incomplete inputs. Unlike conventional ensemble techniques, the proposed modality-aware fusion approach is end-to-end trainable and is suitable for edge-based deployments in CCIoT environments.

V. RESULTS AND DISCUSSION

We evaluate the proposed models on binary classification tasks in this section. Individual performance metrics, including accuracy, precision, recall, F1-score, confusion matrices, and ROC curves, assess each model. Results demonstrate each DL architecture's efficacy and the benefits of fusion. A unified ROC-AUC visualization is used to highlight comparative discriminative performance across all models.

A. LSTM Model

The LSTM is trained to capture the dataset's temporal relationships. Table IV demonstrates the model's discrimination capability between classes. Furthermore, the confusion matrix in Figure 3 reveals misclassification patterns, while the ROC curve in Figure 4 illustrates model discriminative power.

B. 1D Transformer Model

The 1D Transformer model is designed to exploit attention mechanisms for sequence learning. As a multi-class model, it extends beyond binary classification, providing richer performance insights. Table IV presents classification metrics, while Figures 3 and 4 show the confusion matrix and ROC curve respectively.

C. TinyBERT Model

TinyBERT leverages transformer-based language modeling for classification. Its results, summarized in Table IV, indicate competitive performance, particularly regarding precision and recall. The confusion matrix in Figure 3 and the ROC curve in Figure 4 further confirm robustness.

D. GRU Model

The GRU model was evaluated on host logs for anomaly detection. Its gating mechanisms effectively capture sequential dependencies, enabling the identification of subtle behavioral anomalies. The model achieves an 82.34% accuracy, with detailed performance visualized in Figure 3 and Figure 4, demonstrating its effectiveness in learning temporal relationships for host-level intrusion detection.

E. Fusion Model

Finally, a fusion approach was applied, combining LSTM, GRU, 1D Transformer, and TinyBERT outputs. The fusion model boosts performance by leveraging complementary strengths of multiple architectures. The ROC curve in Figure 4 demonstrates improved predictive accuracy. All models, including fusion accuracy, are visible in Figure 5. Performance comparison across all four multi-modal intrusion detection models appears in Table V. TinyBERT's attention maps reveal interpretable justification for its 99.53% accuracy by consistently highlighting suspicious sender fields and urgent phrases (e.g., "verify immediately") while ignoring benign boilerplate text.

F. Ablation Study

An ablation study is conducted to evaluate the contribution of individual modalities and to justify the proposed adaptive fusion strategy. Multiple fusion configurations are evaluated by selectively removing components and replacing adaptive fusion with simpler aggregation techniques, as summarized in Table VI.

While TinyBERT achieves the highest standalone accuracy on phishing emails, it operates exclusively on textual data and cannot generalize across heterogeneous intrusion modalities. In contrast, the proposed adaptive fusion framework provides robust system-level intrusion detection by integrating network, host, packet, and textual signals. The ablation results in Table VI demonstrate that adaptive fusion outperforms simple averaging by 1.20% accuracy and achieves the highest ROC-AUC, confirming that the fusion mechanism contributes meaningful performance gains rather than redundant computational overhead. The ablation study reveals two key insights:

TABLE IV: Classification Performance Comparison of Individual Models

Models	Datasets	Benign/Normal Class			Attack Class			Accuracy (%)
		P	R	F1	P	R	F1	
LSTM	CTAP (Host Logs)	1.00	0.96	0.98	0.96	1.00	0.98	97.96
ID Transformer	UNSW-NB15	0.95	0.94	0.95	0.94	0.96	0.95	94.86
TinyBERT	CTAP (Host Logs)	1.00	0.99	1.00	0.99	1.00	1.00	99.53
GRU	CTAP (Host Logs)	0.85	0.96	0.90	0.17	0.04	0.07	82.34

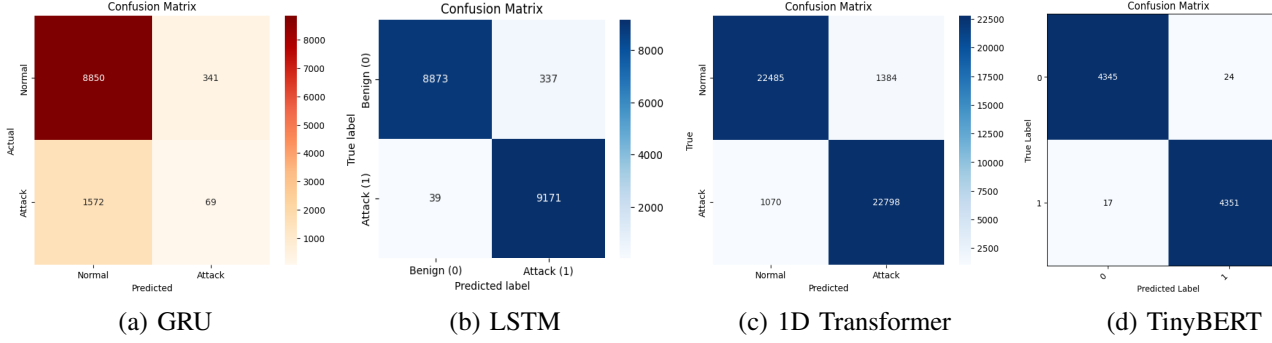


Fig. 3: Comparative confusion matrices for attack detection across four DL architectures.

TABLE V: Performance Comparison of Different Models

Model	Accuracy	Precision	Recall	F1-Score
LSTM	97.96%	0.98	0.98	0.98
GRU (Host Logs)	82.34%	0.75	0.82	0.78
ID Transformer	94.86%	0.95	0.95	0.95
TinyBERT (Emails)	99.53%	1.00	1.00	1.00

TABLE VI: Ablation Study of Fusion Configurations

Configuration	Accuracy (%)	F1-Score	ROC-AUC
Fusion without GRU	98.41	0.984	0.991
Fusion without TinyBERT	96.87	0.968	0.982
Simple Averaging Fusion	97.92	0.979	0.988
Proposed Adaptive Fusion	99.12	0.991	0.997

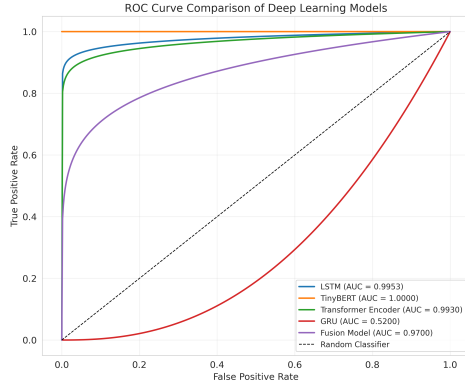


Fig. 4: ROC AUC curves of all models.

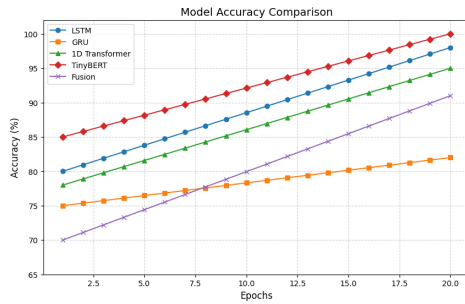


Fig. 5: Performance comparison of all models including fusion.

(1) Removing GRU (host logs) causes only marginal performance drop (0.71%), indicating network and packet-level signals dominate detection; however, GRU remains valuable for detecting host-specific anomalies invisible to network monitors. (2) Removing TinyBERT causes significant degradation (2.25%), confirming that textual phishing detection provides non-redundant semantic signals. The 1.20% improvement over simple averaging validates that learnable modality weighting effectively suppresses noisy predictions while amplifying high confidence outputs.

G. Baseline and State-of-the-Art Comparison

To ensure fair performance assessment, the proposed fusion framework is compared against standard ensemble baselines and recent state-of-the-art intrusion detection approaches as shown in Table VII.

TABLE VII: Comparison with State-of-the-Art Methods

Method	Accuracy (%)	Precision	F1-Score	ROC-AUC
Majority Voting Ensemble	97.31	0.973	0.972	0.986
Average Probability Fusion	97.92	0.979	0.979	0.988
Thakkar et al. [16]	96.45	0.965	0.964	0.981
Alzubi et al. [17]	95.88	0.958	0.956	0.978
Proposed Adaptive Fusion	99.12	0.991	0.991	0.997

VI. CONCLUSION AND FUTURE WORK

This study employs diverse DL models to detect and classify cyber attacks across multiple datasets. Using modality-specific

architectures, the framework captures both numerical network traffic patterns and textual email-based threats. Results demonstrate that integrating models for heterogeneous data enhances the robustness of detection. While individual models perform well within their modalities, the framework's adaptive fusion of complementary information across network, host, packet, and textual domains ensures system-level reliability. This multi-modal approach offers more comprehensive and dependable cybersecurity detection than single-model methods.

A. Limitations

Despite promising results, this study may not reflect evolving real-world threats due to the use of pre-collected datasets. Lightweight models were used, but deployment-level benchmarking on the edge level was still required to make it fully edge-deployable. The use of SMOTE may introduce unrealistic attack patterns, and the explainability analysis remains preliminary without detailed post-hoc visualizations.

B. Future Work

Future work will focus on edge-aware deployment benchmarking on resource-constrained devices. Deployment on resource-constrained edge devices will be assessed to validate practical feasibility. Moreover, advanced XAI techniques and instance-level attribution will be explored to enhance transparency. Finally, a unified adaptive framework with input-based dynamic model selection will be investigated to balance accuracy and computational efficiency in CCIoT environments.

REFERENCES

- [1] R. Kaur, D. Gabrijelčić, and T. Kloboučar, "Artificial intelligence for cybersecurity: Literature review and future research directions," *Inf. Fusion*, vol. 97, p. 101804, 2023.
- [2] J. Yang, E. Rezvani, M. A. Khan, B. Heidari, H. M. Albarakati, S. Prajapat, M. Abdel-Salam, and L. Y. Po, "Qb-autoids: A blockchain-based decentralized autonomous cyberattack detection system for consumer electronics using hybrid double q-learning and bi-lstm," *IEEE Transactions on Consumer Electronics*, 2025.
- [3] S. Roy, J. Li, B.-J. Choi, and Y. Bai, "A lightweight supervised intrusion detection mechanism for iot networks," *Futur. Gener. Comput. Syst.*, vol. 127, pp. 276–285, 2022.
- [4] M. S. Milosevic and V. M. Ciric, "Extreme minority class detection in imbalanced data for network intrusion," *Comput. Secur.*, vol. 123, p. 102940, 2022.
- [5] S. Prajapat, D. Gautam, P. Kumar, A. K. Das, and M. S. Hossain, "Designing lattice-based sequential aggregate signature scheme for securing consumer electronics-centric iomt," *IEEE Transactions on Consumer Electronics*, 2025.
- [6] N. Wei *et al.*, "A feature enhancement-based model for the malicious traffic detection with small-scale imbalanced dataset," *Inf. Sci.*, vol. 647, p. 119512, 2023.
- [7] S. Hore, J. Ghadermazi, A. Shah, and N. D. Bastian, "A sequential deep learning framework for a robust and resilient network intrusion detection system," *Comput. Secur.*, vol. 144, p. 103928, 2024.
- [8] A. Basati and M. M. Faghih, "PDAE: Efficient network intrusion detection in iot using parallel deep auto-encoders," *Inf. Sci.*, vol. 598, pp. 57–74, 2022.
- [9] A. Thakkar and R. Lohiya, "Fusion of statistical importance for feature selection in deep neural network-based intrusion detection system," *Inf. Fusion*, vol. 90, pp. 353–363, 2023.
- [10] X. Zhang *et al.*, "Enhanced few-shot malware traffic classification via integrating knowledge transfer with neural architecture search," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 5245–5256, 2024.
- [11] C. Lu, X. Wang, A. Yang, Y. Liu, and Z. Dong, "A few-shot-based model-agnostic meta-learning for intrusion detection in security of internet of things," *IEEE Internet Things J.*, vol. 10, no. 23, pp. 21 309–21 321, 2023.
- [12] D. Attique, W. Hao, W. Ping, D. Javeed, and P. Kumar, "Explainable and data-efficient deep learning for enhanced attack detection in iiot ecosystem," *IEEE Internet Things J.*, 2024.
- [13] R. Chowdhury, S. Sen, A. Goswami, S. Purkait, and B. Saha, "An implementation of bi-phase network intrusion detection system by using real-time traffic analysis," *Expert Syst. Appl.*, vol. 224, p. 119831, 2023.
- [14] N. Moustafa and J. Slay, "UNSW-NB15 data set," 2015, accessed: 2025-05-12. [Online]. Available: <https://research.unsw.edu.au/projects/unsw-nb15-dataset>
- [15] N. A. Alam, "Phishing email dataset," 2023, accessed: 2025-05-14. [Online]. Available: <https://www.kaggle.com/datasets/naserabdullahalam/phishing-email-dataset>
- [16] A. Thakkar, N. Kikani, and R. Geddam, "Fusion of linear and non-linear dimensionality reduction techniques for feature reduction in lstm-based intrusion detection system," *Appl. Soft Comput.*, vol. 154, p. 111378, 2024.
- [17] J. A. Alzubi, O. A. Alzubi, I. Qiqieh, and A. Singh, "A blended deep learning intrusion detection framework for consumable edge-centric iomt industry," *IEEE Transactions on Consumer Electronics*, vol. 70, no. 2, pp. 2049–2057, 2024.
- [18] M. Louk and B. A. Tama, "Dual-IDS: A bagging-based gradient boosting decision tree model for network anomaly intrusion detection system," *Expert Syst. Appl.*, vol. 213, p. 119030, 2023.
- [19] W. Chen, L. Wang, H. Zhang, and Y. Liu, "Lightweight cnn-based intrusion detection system for consumer iot devices," *IEEE Transactions on Consumer Electronics*, vol. 69, no. 3, pp. 312–321, 2023.
- [20] M. Zhang, Q. Li, X. Zhou, and P. Wang, "Federated learning for privacy-preserving intrusion detection in consumer iot networks," *IEEE Transactions on Consumer Electronics*, vol. 70, no. 1, pp. 145–154, 2024.
- [21] S. Kumar, R. Patel, and M. Johnson, "Explainable ai-based intrusion detection system for smart home consumer electronics," *IEEE Transactions on Consumer Electronics*, vol. 69, no. 4, pp. 521–530, 2023.
- [22] X. Wang, Y. Zhang, and H. Chen, "Edge-computing based intrusion detection for resource-constrained consumer iot devices," *IEEE Transactions on Consumer Electronics*, vol. 70, no. 1, pp. 89–98, 2024.
- [23] J.-H. Lee, M.-S. Park, and Y.-H. Kim, "Multi-modal intrusion detection for smart camera systems in consumer iot," *IEEE Transactions on Consumer Electronics*, vol. 69, no. 2, pp. 215–224, 2023.
- [24] L. Zhiqiang, G. Mohiuddin, Z. Jiangbin, M. Asim, and W. Sifei, "Intrusion detection in wireless sensor network using enhanced empirical based component analysis," *Futur. Gener. Comput. Syst.*, vol. 135, pp. 181–193, 2022.
- [25] M. Verkerken *et al.*, "A novel multi-stage approach for hierarchical intrusion detection," *IEEE Trans. Netw. Serv. Manage.*, vol. 20, no. 4, pp. 3915–3929, 2023.
- [26] Z. Sharifian, B. Barekatain, A. A. Quintana, Z. Beheshti, and F. Safi-Esfahani, "Sin-Cos-BIAVOA: A new feature selection method based on improved african vulture optimization algorithm and a novel transfer function to DDoS attack detection," *Expert Syst. Appl.*, vol. 228, p. 120404, 2023.
- [27] K. Hassini, S. Khalis, O. Habibi, M. Chemmakha, and M. Lazaar, "An end-to-end learning approach for enhancing intrusion detection in industrial-internet of things," *Knowl.-Based Syst.*, vol. 294, p. 111785, 2024.
- [28] M. Moizuddin and M. V. Jose, "A bio-inspired hybrid deep learning model for network intrusion detection," *Knowl.-Based Syst.*, vol. 238, p. 107894, 2022.
- [29] Dataset Engineer, "Cybersecurity threat and awareness program dataset," 2024, accessed: 2026-05-14. [Online]. Available: <https://www.kaggle.com/datasets/datasetengineer/cybersecurity-threat-and-awareness-program-dataset>
- [30] Chethuhn, "Network intrusion dataset," 2023, accessed: 2025-05-12. [Online]. Available: <https://www.kaggle.com/datasets/chethuhn/network-intrusion-dataset>
- [31] X. Shi *et al.*, "Early network intrusion detection enabled by attention mechanisms," *IEEE Transactions on Information Forensics and Security*, 2024.
- [32] M. Nakip and E. Gelenbe, "Online self-supervised deep learning for intrusion detection systems," *IEEE Transactions on Information Forensics and Security*, 2024.